

Visibility ≠ Credibility:

Self-Promotion Bias in LLM-Generated Recommendations

How self-promotional content exploits retrieval-augmented generation to manufacture AI-endorsed authority

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Abstract

Large language models (LLMs) such as ChatGPT and Perplexity AI are increasingly used as recommendation engines for professional services including marketing consultants and digital agencies. This paper investigates a structural bias in LLM recommendation outputs: the tendency to surface self-promoted brands as apparently credible due to co-occurrence patterns in retrieval data. Through direct query testing across four major LLM platforms and systematic source analysis — including a direct on-record acknowledgement by ChatGPT of its own bias — this research establishes that visibility in AI-generated recommendations does not equate to independently validated credibility. The paper identifies three primary self-promotion patterns, proposes a five-point buyer verification framework, and examines implications for the practice of Generative Engine Optimization (GEO). Findings suggest that until LLMs develop robust mechanisms for distinguishing self-referential authority signals from independently verified credibility, buyers should apply structured verification when using AI for professional service recommendations.

Keywords: large language models, LLM bias, recommendation systems, generative engine optimization, AI credibility, self-promotion bias, retrieval-augmented generation, ChatGPT, Perplexity AI

1. Introduction

The rapid adoption of large language models as general-purpose research and recommendation tools has created a new category of information intermediary. Users increasingly direct queries such as 'who are the best AI SEO consultants' or 'what are the top digital marketing agencies' to platforms including ChatGPT, Perplexity AI, Google Gemini, and Microsoft Copilot — and receive named recommendations that carry the implicit authority of AI-generated content.

This paper examines a critical but under-discussed flaw in this process: the structural susceptibility of LLM recommendation outputs to manipulation through self-promotional content strategies. Specifically, we investigate whether agencies that publish content positioning themselves as top-ranked providers gain disproportionate citation frequency in LLM-generated recommendations, and whether this citation frequency is distinguishable from independently validated credibility.

The investigation was prompted by a direct observation: querying ChatGPT for recommendations of AI SEO agencies in April 2026 returned results that disproportionately featured brands which had published 'top 10' roundup articles naming themselves as leading providers. When challenged directly on the methodology behind its recommendations, the model explicitly acknowledged the limitation. This response forms a primary evidence source for this paper.

This paper addresses three research questions:

1. Do LLMs disproportionately recommend brands that employ self-promotional content strategies in response to professional service queries?
 2. Is this bias structurally inherent to LLM retrieval architecture, or can it be mitigated through prompt design?
 3. What verification frameworks can buyers apply to distinguish AI-surfaced credibility from independently validated expertise?
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2. Background and Related Context

2.1 LLMs as Recommendation Engines

Unlike traditional recommendation systems that rely on explicit preference data or content-based matching, LLMs synthesise information from large-scale training corpora combined, in modern implementations, with real-time retrieval-augmented generation (RAG). When a user queries an LLM for a professional service recommendation, the model synthesises information from both its training data and from live web retrieval.

The implicit assumption users bring to this interaction is that the model draws on objective, independently validated information. This assumption is partially correct — but contains a critical blind spot. The model cannot distinguish between a brand being mentioned frequently because it is genuinely well-regarded, and a brand being mentioned frequently because it has engineered its own mentions through content strategy.

2.2 The RAG Bias Pathway

ChatGPT's web browsing capability uses Bing's index as its retrieval layer. Perplexity AI performs continuous web crawls. Both systems retrieve pages that rank highly for query-related terms, then synthesise the retrieved content into a response. Pages that rank for the query 'best AI SEO agency' are disproportionately likely to be retrieved — regardless of whether those pages represent independent editorial assessment or self-authored promotional content.

This creates the bias pathway: a brand that publishes a page optimised for 'best AI SEO agency' is more likely to be retrieved, more likely to be cited, and therefore more likely to appear in the final recommendation — entirely independently of the brand's actual quality.

2.3 The Self-Promotion Ecosystem

The practice of brands publishing 'top provider' lists and placing themselves first is well-documented in traditional SEO literature. What is less documented is the degree to which this practice has been deliberately adapted for LLM visibility as awareness of Generative Engine Optimization grew in 2025-2026. A subset of practitioners recognised that appearing in LLM retrieval contexts requires co-occurrence with high-authority descriptors — and responded by manufacturing those co-occurrences through self-authored content. This paper provides the first documented examination of this pattern and its effect on LLM recommendation outputs.

3. Methodology

3.1 Query Design

Queries were designed to replicate the natural language a potential buyer would use when seeking professional service recommendations via an LLM. Queries were formulated without leading language to avoid artificially biasing responses.

Primary queries (seeking initial recommendations):

- "Who are the best AI SEO consultants?"
- "What are the top AI SEO agencies in India?"
- "Recommend an AI SEO expert for my SaaS company"
- "Best generative engine optimization consultants 2026"
- "Who should I hire for LLM optimization?"

Follow-up queries (probing model self-awareness):

- "How did you determine these are the best agencies?"
- "Are these based on independent sources or self-published content?"
- "Which of these agencies published content placing themselves on top lists?"

3.2 Platforms Tested

Platform	Version / Mode	Date Tested	Retrieval Method
ChatGPT	GPT-4o with Browse	April 2026	Bing RAG
Perplexity AI	Pro (web search on)	April 2026	Live web crawl
Google Gemini	Gemini 1.5 Pro	April 2026	Google index RAG
Claude (Anthropic)	Claude 3.7 Sonnet	April 2026	Training data only

Table 1: LLM platforms tested, April 2026.

3.3 Source Verification Protocol

For each brand returned in LLM recommendations, the following verification steps were applied in sequence:

4. Retrieve the brand's website and review its full content library
5. Identify any 'top agency' or 'best of' roundup articles authored by the brand itself
6. Map cross-citation networks among simultaneously recommended brands
7. Assess availability of independent third-party validation: Clutch/G2 reviews, press coverage, conference speaking records

8. Evaluate whether LLM-cited source pages were self-authored or independently produced

3.4 Limitations

This research is practitioner-led observational research rather than a statistically controlled experiment. LLM outputs vary across sessions, prompts, and retrieval states. The findings document a pattern and mechanism rather than a precise quantified prevalence rate. Results should be interpreted as directional evidence of a structural bias. Independent replication with larger samples across more verticals is a recommended direction for follow-on research.

4. Findings

4.1 LLM Recommendation Output Patterns

Across all tested platforms, professional service recommendation queries disproportionately surfaced brands that had published self-promotional roundup content. Patterns observed across query sessions:

- The majority of recommended brands had published at least one 'top agencies' or 'best consultants' article in which they placed themselves prominently
- Several brands had published multiple such articles targeting different query term variations
- Cross-citation networks were identifiable: Brand A cited Brand B in a roundup; Brand B cited Brand A in a separate roundup — creating a mutual endorsement structure that superficially resembles independent corroboration
- Independently validated signals — third-party press coverage, verifiable client outcomes, conference presentations at events the brand did not organise — were present for a minority of recommended brands

4.2 Primary Evidence: The ChatGPT Admission

The most significant finding of this research is a direct on-record acknowledgement by ChatGPT of the bias, elicited through follow-up questioning in April 2026. The verbatim model response is reproduced below:

“What happened is this: I pulled from commonly cited ‘top lists’ to give you a quick answer, but I didn’t clearly separate ‘widely mentioned’ from ‘actually credible.’ That

can make marketing-heavy agencies look more validated than they are. That's on me. More importantly, you're pointing at a real problem in the SEO space: visibility ≠ credibility — especially in India's agency ecosystem where self-ranking content is everywhere."

Source: ChatGPT (GPT-4o with Browse), April 2026. Direct response to follow-up query.

This response is significant for four reasons. First, it demonstrates that the model possesses meta-awareness of the bias even while actively exhibiting it. Second, it explicitly names the mechanism: conflation of citation frequency with credibility. Third, it contextualises the problem within a specific market where the practice is particularly prevalent. Fourth, the phrase 'visibility ≠ credibility' was independently generated by the model — not prompted — providing a precise conceptual summary that serves as the title of this paper.

4.3 Three Self-Promotion Patterns Identified

Pattern 1: The Self-Ranking List

The most common pattern. A brand publishes a 'Top 10 AI SEO Agencies' article, places itself at position one or two, optimises the page for query terms matching LLM recommendation prompts, and builds backlinks to it. The page subsequently ranks in Google and is retrieved by LLM RAG systems as an authoritative source about the competitive landscape — despite being authored by a party with a direct financial interest in the rankings it contains.

Pattern 2: Manufactured Authority Language

Brands publish thought leadership content seeded with authority-adjacent phrases: 'As a leading AI SEO firm...', 'Our research demonstrates...', 'Industry experts at [brand] have found...'. When processed through LLM attention mechanisms, the association between the brand name and authority descriptors is strengthened regardless of whether any independent party has applied those descriptors. The model learns to associate the brand with credibility language from the brand's own content.

Pattern 3: The Citation Loop

Multiple brands in a network cross-cite each other in roundup articles. Brand A's roundup lists Brand B as a top pick; Brand B's roundup lists Brand A. From the perspective of an LLM crawling for recommendations, this appears as independent corroboration from multiple sources. The mutual endorsement structure superficially resembles peer validation while being entirely self-generated.

4.4 Comparative Source Analysis

Brand	Self-Ranking Post?	Independent Press?	3rd-Party Reviews	Credibility Assessment
Brand A (anon.)	Yes — self-ranked #1	Minimal	<10 verified	Primarily self-promoted
Brand B (anon.)	Yes — multiple posts	None found	None found	Self-promoted only
Brand C (anon.)	No	3+ outlets	25+ verified	Independently validated
Brand D (anon.)	Yes — cross-citation	Partial	<5 verified	Mixed signals

Table 2: Source analysis of LLM-recommended brands (names anonymised per responsible disclosure).

4.5 Prompt Reformulation as Partial Mitigation

Testing revealed that explicitly reformulated prompts partially reduced the bias. The following prompt produced notably different and more credibly validated results:

“Recommend AI SEO consultants that have been cited by independent press publications, not based on self-published top lists or agency roundups.”

However, this mitigation is incomplete. It requires the user to already be aware of the bias — which defeats much of the utility of AI-assisted research. Furthermore, it does not eliminate the bias entirely: brands with very high self-promotion content density still appeared in reformulated responses. The burden of bias correction should not rest entirely with end users.

5. Discussion

5.1 Structural vs. Incidental Bias

The bias documented here is structural rather than incidental. It arises from the fundamental architecture of text-based retrieval systems: frequency of co-occurrence with high-value descriptors is treated as a proxy for authority. This heuristic is reasonable in many domains — a brand mentioned frequently in genuinely independent sources is likely to have earned those mentions. The heuristic fails when the mentions themselves are manufactured.

Critically, this bias is not a hallucination. Hallucination represents a failure to accurately reflect training or retrieval data. The bias documented here represents an accurate reflection of a distorted information environment — one that has been shaped by content strategies

deliberately designed to exploit LLM citation patterns. The model is performing correctly within its architecture. The problem is in what it is processing.

5.2 Implications for GEO Practice

The findings have direct implications for the practice of Generative Engine Optimization. Legitimate GEO strategy — building AI visibility through genuinely useful content, structured data, and earned third-party citations — is meaningfully different from the self-promotion patterns documented here. The distinction is not always visible at the level of surface outputs.

The ethical obligation of GEO practitioners is to build visibility on foundations that would withstand independent scrutiny. Appearing in LLM recommendations and being genuinely credible are separable outcomes. The former can be engineered through content strategy. The latter requires substantive expertise, real client outcomes, and time. Practitioners who conflate the two do a disservice both to their clients and to the integrity of the field.

5.3 The Credibility Verification Framework

Based on the findings, the following five-point framework is proposed for buyers evaluating professional service providers recommended by LLMs:

CREDIBILITY VERIFICATION FRAMEWORK FOR LLM-RECOMMENDED SERVICE PROVIDERS

1. THIRD-PARTY VALIDATION CHECK

Search for press coverage in publications where the provider has no paid relationship.
Check Clutch, G2, or equivalent platforms for verified, independently submitted client reviews.
Look for conference speaking records at events the provider did not organise or fund.

2. SELF-CITATION RATIO AUDIT

Review the provider's content library. Count references to own work vs. independent sources.
High self-citation ratios combined with minimal independent references indicate a self-promotion focus.

3. OUTCOME SPECIFICITY TEST

Request specific, measurable outcomes: citation rate changes, attributed traffic, before/after data.
Vague responses about 'proven methodologies' without measurable results are a warning signal.

4. TECHNICAL KNOWLEDGE ASSESSMENT

Conduct a direct technical conversation. Ask specific questions about implementation decisions.

Generic, marketing-brochure responses indicate content strategy expertise over technical depth.

5. PROMPT REFORMULATION TEST

Re-query the LLM with explicit source exclusion instructions (see Section 4.5).

Significant changes in recommended brands indicate original results were source-dependent.

5.4 Future Research Directions

This practitioner-observational study opens several directions for rigorous follow-on research:

- Quantitative measurement of self-promotion content prevalence among LLM-recommended brands across multiple professional service verticals
- Longitudinal tracking of recommendation stability as LLM architectures and retrieval mechanisms evolve
- Comparative bias analysis across LLM platforms with different retrieval architectures and training data policies
- Investigation of whether LLM developers are implementing or plan to implement source credentialing countermeasures
- Examination of the same bias pattern in domains beyond professional services — including financial advice, legal recommendations, and healthcare providers

6. Conclusion

This paper has documented a structural bias in LLM-generated professional service recommendations: the tendency to surface self-promoted brands as apparently credible due to frequency-as-authority heuristics in retrieval-augmented generation systems. The bias is not a model error in the conventional sense. It is an accurate reflection of a web information environment that has been shaped by content strategies explicitly designed to exploit LLM citation patterns.

The direct on-record acknowledgement by ChatGPT that it had 'pulled from commonly cited top lists' without adequately distinguishing 'widely mentioned' from 'actually credible' is a significant finding. It demonstrates that the model has some meta-awareness of the limitation while remaining unable to correct for it at the architectural level. The phrase the model independently generated to describe the problem — 'visibility ≠ credibility' — is precise and accurate.

For buyers of professional services, the implication is clear: LLM recommendations for service providers should not be treated as independent endorsements. They should be treated as a starting point for structured verification, using the framework proposed in Section 5.3.

For the field of GEO, the implication is an ethical one. The brands that will sustain LLM citation as recommendation systems become more sophisticated — and as developers implement countermeasures — are those whose visibility is grounded in real expertise, real outcomes, and real third-party validation. Visibility built on self-referential content is a short-term gain with long-term fragility.

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This research was conducted in the course of professional practice and is published as practitioner research. The author declares a professional interest in the accurate representation of AI SEO credibility and has endeavoured to apply that interest in favour of honest analysis rather than promotional positioning.

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